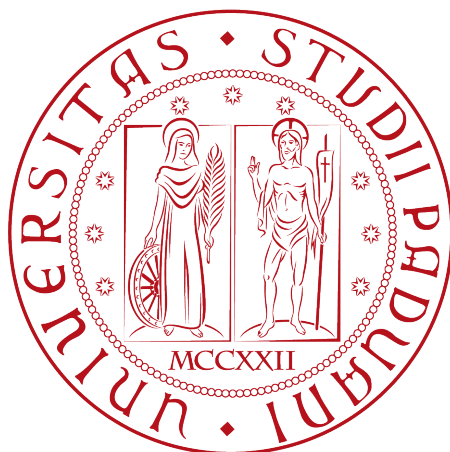


University of Padua

DEPARTMENT OF MATHEMATICS “TULLIO LEVI-CIVITA”

MASTER DEGREE IN COMPUTER SCIENCE



GeoMedia: A Geographic Framework for Location-Based Content Sharing

Master's Thesis

Supervisor

Prof. Claudio Enrico Palazzi

Co-supervisor

Dott. Lorenzo Perinello

Candidate

Alberto Morini

Student ID

2107783

ACADEMIC YEAR 2025-2026

Summary

Over the past few decades, the use of mobile devices has increased drastically, almost replacing desktop software and web browsers. This shift has been driven not only by the widespread adoption of smartphones, but also by the availability of integrated sensors such as GPS, gyroscopes, and Bluetooth, which allows a richer and more context-aware user experiences. By leveraging these technologies, Online Social Media platforms have been able to provide more advanced and sophisticated features; consequently collecting larger amounts of metadata, interaction and daily engagement.

This thesis presents “GeoMedia”, an open-source framework designed for sharing multimedia content — such as text, images, audio, and files — anchored to real-world geographic locations. By leveraging GPS data available on mobile devices, the application provides an interactive map through which users can explore and publish digital content, creating a seamless integration between the digital and physical worlds. Moreover, the framework offers several features and cover various use cases, including restricting content visibility within customizable geographic areas, enabling time-sensitive content, applying user-based restrictions for content creation and visualization, and supporting sequential visibility between published posts. Furthermore, analyzes the motivations behind GeoMedia, its critical aspects and design considerations, describes its system architecture and implementation through a comparison of different technological paradigms. Finally, it explores potential applications and outlines possible future developments of the framework.

An previous version of this framework has been presented at “GoodIT 2025: International Conference on Information Technology for Social Good” (September 3 - 5, 2025 - Antwerp, Belgium) [1]

Contents

1	Introduction	1
1.1	Overview	1
1.2	Online Social networks	2
2	State of Art	3
2.1	Foursquare Swarm	3
2.1.1	Dodgeball and Google Latitude	3
2.2	Snapchat	4
2.3	Instagram	5
2.3.1	Instagram Map	5
2.4	Google Maps	6
2.5	Jagat	6
2.6	Others	6
3	GeoMedia	7
3.1	Idea	7
3.2	Functionalities	7
4	Architecture and Software design	8
5	Mobile app	9
6	Geomedia Backend	10
7	Conclusions	11
7.1	Consuntivo finale	11
7.2	Raggiungimento degli obiettivi	11
7.3	Conoscenze acquisite	11
7.4	Valutazione personale	11
A	Appendice A	13
	Acronyms and abbreviations	14
	Glossary	15
	Bibliography	16

List of Figures

2.1	Snapchat functionalities	4
2.2	Snapchat ‘my place’ functionality	4
2.3	Instagram map discovery	6
2.4	Instagram map content 2026	6

List of Tables

List of Codes

Chapter 1

Introduction

1.1 Overview

Technology has always influenced society, from the first introduction to the Personal Computer to the Internet, changing the world year after year, from personal life to workspaces, people are surrounded by software on their computers and mobile devices. Over the last decades, the investments in the development of technological innovations, both in hardware and software, as increased exponentially, shaping economical and social impact by integrating the latest features into everyday life. Smartphones and tablets have been the core engine that completely changed daily habits and digital usage; their wide adoption has reached almost three quarters of global population [44, 39], making users abandon traditional computers, not just for web browsing but more especially for social media[38] and instant messaging platform[41], that changed the design and development of applications into a ‘mobile first’ logic.

Compared to computers, mobile devices offer distinct functionalities and strengths that have facilitated their widespread adoption. These can be categorized according to the fundamental logic underlying any technology:

1. **Hardware:** the increase in computational performance[26], battery life[6], fast connectivity, and the integration of several built-in sensors[25] such as gyroscopes, GPS, NFC, light sensors, cameras, and vibration motors. These enable applications to interact with different fields such as health[4], education[31], and, through motion-sensors or cameras, an interaction with physical world, creating more immersive user experience.
2. **Software:** the development and presence of mobile applications has played a crucial role. In fact, platforms such as ‘Windows Phone’ were eventually abandoned and discontinued due to the lack of applications [5, 45] compared to operating systems like ‘iOS’[12] and ‘Android’[13].

In this way, smartphones apps have become a key integration with daily life and physical world, for example, when planning a trip or just driving, smartphones, integrated with cars assists users via app like ‘Google Maps’[15] or ‘Waze’[10], providing not just directions but also real-time traffic updates, read events or information about available parking. This makes the surrounding environment enriched with digital details. Another example is ‘Google Lens’[14], which can identify objects from a photograph, or ‘Shazam’[37] that through devices’ microphone can recognize a playing song. Even in workplaces or public areas, technology integrations can be present, from the restaurant that provide a digital menu by a tablet, to customers checking the nutrition macros into a grocery store by scanning the barcode on the pack, to museums by scanning a QR Code enabling audio-guided tours or unlock Augmented Reality functionalities.

1.2 Online Social networks

Alongside the widespread adoption of smartphones and mobile devices, online social networks, have emerged as one of the most influential and important sectors of the technology market. Today, social media platforms rank among the firsts most used applications worldwide [32, 11], while the companies that provide them become some of the most bigger actors in global financial markets [33, 22, 18].

Online Social Networks (OSN) are the digitalized form of a Social Network (typically represented as a graph of people where links correspond to relationships) through the Internet[34]. For instance, Facebook, one of the largest platforms, represents a graph of friendships, where nodes (users) are connected via undirected edges, assuming that friendships are mutual. There are also different types of OSN, such as Instagram or Twitter, where the links between users are directed, implementing a “following” model.

These services provide a variety of functionalities, first of all instant messaging capabilities, both one-to-one and one-to-many, and, in addition, the ability to share a wide range of multimedia content, including images, videos, audio recordings, and other types of documents. This facility and advantages compared to other communication forms, like SMS, has led to their integration across diverse social, professional, and public contexts. The particularity of sharing multimedia content, commonly referred to as a ‘post’, has coined the term Social Media, where the published content represent the node of interaction between users through comments or reactions (e.g., ‘like’ or ‘unlike’), thus to replicate human behavior in dialogues and social interactions.

Despite the significant benefits offered by social media services, including enhanced connectivity, community creation, and information sharing, their adoption has also introduced various social and ethical challenges. Concerns regarding user privacy are particularly relevant, as it is often not respected by the service providers themselves, who collect personal information and interests in order to monetize them or enable surveillance practices[23]. Other serious issues include the dissemination of misinformation and fake news, nowadays further amplified by Artificial Intelligence-generated content, commonly known as ‘deepfakes’ [24]. Such content can influence public opinion and social behavior, not always in a positive way [43, 27].

Online Social Networks have also intensified common societal problems through phenomena such as ‘echo chambers’[3], where users are repeatedly exposed to information and content that aligns with their existing beliefs and preferences. This continuous reinforcement can strengthen existing views, reduce exposure to alternative perspectives, and, in some cases, contribute to ideological radicalization. In practice, examples can be observed in so-called ‘incel’ communities [28] and among supporters of extreme political movements.

To address these issues, social media platforms have adopted various moderation and safety mechanisms, including automated content analysis systems, community guidelines and reporting tools. While these measures contribute reducing the spread of harmful content, they are not always fully effective, due to the scale and complexity of emerging innovations, such as ‘deepfake’ [2]. Consequently user reporting and human moderation aside of automated tools, continue to play a crucial role in identifying, evaluating, and removing content that violates policies or legal regulations[8].

Chapter 2

State of Art

GeoMedia positions itself among existing online social media platforms, by offering functionality centered on the sharing of geolocated content linked to specific geographic coordinates. Content is visible only within a configurable distance radius, allowing to create a wide range of real-world applications, including scavenger hunts, interactive city guides, and any other location-based experiences

The platform enables users to discover and interact with the physical environment through a map enriched by content contributed by other users, who can upload pictures or other multimedia resources associated with specific locations.

Although several other social media platforms utilize the GPS capabilities of mobile devices to enhance user experience and provide location-aware features, none fully realize the underlying concept of GeoMedia or offer the same degree of flexibility for creating diverse application scenarios and exploration of uploaded content.

2.1 Foursquare Swarm

Foursquare Swarm is the spin-off of a formerly app, ‘Foursquare City Guide’, which provide personalized recommendations of nearby places to the users’ current location based on its interests or previous browsing history and check-ins (locations previously visited by the user).

Foursquare was discontinued in 2024 and its functionalities have been integrated into Swarm, which in addition allows users to share their position with relatives, maintaining a personal lifelog of visited places, which can be shared with others.

For each visited location, users may leave reviews or upload multimedia content; however, these contributions are strictly tied to physical place, such as stores, cafés, or restaurants. Users cannot publish content at arbitrary geographic coordinates, nor can restrict post visibility, as all content is publicly accessible. The application further encourages the usage through a gamification system, in which users earn badges for visiting locations and uploading content. These mechanisms are broadly comparable to those implemented in other platforms such as Google Maps (see Section 2.4).

2.1.1 Dodgeball and Google Latitude

Dodgeball was a service that simply allowed users to share their current locations with friends through a map, and served as a precursor to Foursquare City Guide and Foursquare Swarm same functionality. Later was acquired by Google and rebranded as ‘Google Latitude’ [17] discontinued in 2013 and integrated into another discontinued social media by Google (Google+ [16]) and finally migrated into Google Maps2.4 in 2018.

2.2 Snapchat

Snapchat [21] has been one of the most used social media platforms worldwide [20], mainly thanks to its instant messaging service and multimedia functionality, such as time-based content, which makes uploaded content visible only for 24 hours or visible only once to a user. This encourages users to check it daily in order not to miss anything and to keep up with relatives' lives. As a negative effect, these features have fueled some social impacts such as FoMO (Fear of Missing Out) [9, 36], leading users to become increasingly attached to the social platform and also creating self-comparison with content posted by other users. This problem is not strictly related to Snapchat, but is very common in every self-centric social media platform.

Snapchat also provides a map-based feature that allows users to view the real-time location of friends and closer content grouped into a unified cluster of information, as illustrated in Figure 2.1. While this functionality differs in structure from GeoMedia Framework 3, it raises important privacy and safety concerns [29].

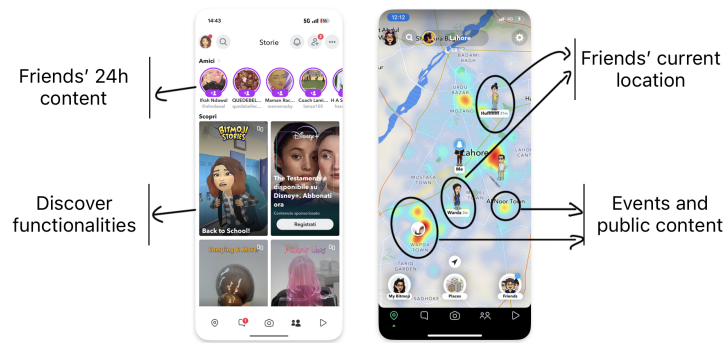


Figure 2.1: Snapchat functionalities

Moreover, as the Swarm app does 2.1, it allows users can log visited locations, including public places such as bars, retail stores, and museums. This enables the association of user-generated content with specific points of interest (POI) and supports the aggregation of geographically and contextually related content to represent events, such as concerts or other real-word gatherings, as illustrated in Figure 2.2

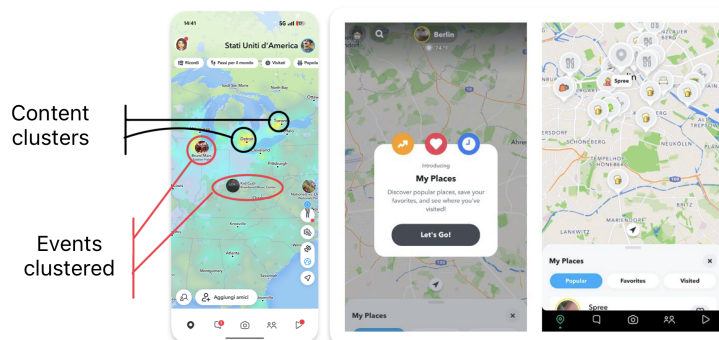


Figure 2.2: Snapchat 'my place' functionality

Snapchat represents a notable example of the adoption of the “mobile-first” paradigm, especially in social media applications. The platform leverages not only GPS module but also a variety of mobile device sensors, including the camera for augmented reality

(AR) experiences and the microphone to records voice-based content. By integrating these hardware features into its core functionality, the app delivers a highly immersive and personalized user experience.

2.3 Instagram

Instagram, owned by Meta Platforms, is one of the biggest and most used social media of the latest decades. Through the introduction of numerous features and continuous innovations, it has attracted more than two billion montly active users worldwide [35, 40].

The platform was initially released in October 2010 exclusively for iOS devices. In fact, at the time, the uploaded content was framed in a square (1:1) aspect ratio with a resolution of 640 pixels, reflecting the display characteristics of iPhones at the time. Later acquired by the formerly named Facebook Inc. (now Meta Platforms Inc.), the application was also deployed for Android systems in April 2012. Then, the original aspect-ratio limitations were removed, and several additional functionalities were introduced, including direct messaging and support for multiple images or videos within a single post.

A major development was the introduction of the “Stories” feature, which provid ethe same mechanism created by Snapchat 2.2, that allows users to publish content that remains visible for only 24 hours after publication. The addition of this ability contributed significantly to Instagram surpassing Snapchat in user engagement metrics by 2019 [42].

To enrich the media sharing, Instagram provides a variety of creative tools, including photographic filters, image-editing capabilities, and even a live-streaming functionality. The platform also popularized the use of hashtags as a content categorization mechanism, enabling users to discover media and connect with communities centered around shared interests and topics.

Instagram to encourage content discovery adopt a sophisticated recommendation systems that leverage contextual information, including geographic location derived from GPS or cellular network data. Such metadata_g contributes to the generation of personalized recommendations, allowing users to discover nearby profiles, locations, and content that align with their interests [19].

Despite its popularity, Instagram has been the subject of significant criticism. Concerns have been raised regarding user privacy, large-scale behavioral profiling, and the potential influence of recommendation algorithms on user behavior. Additional controversies involve the platform’s impact on adolescent mental health, content moderation practices, and the dissemination of illegal or inappropriate content.

2.3.1 Instagram Map

Despite its numerous functionalities, Instagram introduced in 2022 a map-based content discovery feature[7]. By anchoring content to existing geographic entities such as cities or others POIPoint of Interest (POI), users can explore both nearby or remote content, as illustrated in Figure 2.3.

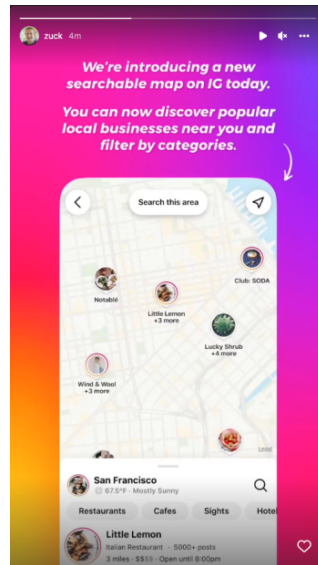


Figure 2.3: Instagram map discovery

In 2026, this functionality underwent a redesign: content is no longer presented as spatial clusters on the map (as implemented in Snapchat, Figure 2.2), instead is represented in a grid-based layout when a specific place is selected via the searchbar of the “Explore” section, as illustrated in Figure 2.4. In addition, Instagram provides functionality for sharing real-time location with users connections (followers), similarly to other social media platforms discussed in Sections 2.2 and 2.1.

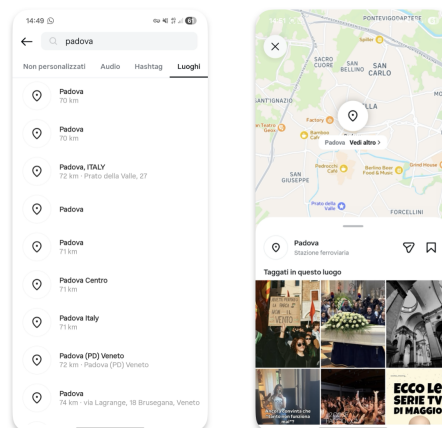


Figure 2.4: Instagram map content 2026

2.4 Google Maps

2.5 Jagat

2.6 Others

- auglinn - wallame?? - strava

Chapter 3

GeoMedia

3.1 Idea

3.2 Functionalities

Chapter 4

Architecture and Software design

Chapter 5

Mobile app

Chapter 6

Geomedia Backend

Chapter 7

Conclusions

7.1 Consuntivo finale

7.2 Raggiungimento degli obiettivi

7.3 Conoscenze acquisite

7.4 Valutazione personale

“ ”

—

Acknowledgements

Innanzitutto, vorrei esprimere la mia gratitudine al Prof. Claudio Enrico Palazzi, relatore della mia tesi, per l'aiuto e il sostegno fornitomi durante la stesura del lavoro.

...

...

Padua, May 2026

Alberto Morini

Appendix A

Appendice A

Citazione

Autore della citazione

Acronyms and abbreviations

API [Application Program Interface](#). [15](#)

POI Points of Interest or Point of Interest. [5](#)

Glossary

API in informatica con il termine *Application Programming Interface API* (ing. interfaccia di programmazione di un'applicazione) si indica ogni insieme di procedure disponibili al programmatore, di solito raggruppate a formare un set di strumenti specifici per l'espletamento di un determinato compito all'interno di un certo programma. La finalità è ottenere un'astrazione, di solito tra l'hardware e il programmatore o tra software a basso e quello ad alto livello semplificando così il lavoro di programmazione.
14

Metadata in informatica con il termine *Application Programming Interface API* (ing. interfaccia di programmazione di un'applicazione) si indica ogni insieme di procedure disponibili al programmatore, di solito raggruppate a formare un set di strumenti specifici per l'espletamento di un determinato compito all'interno di un certo programma. La finalità è ottenere un'astrazione, di solito tra l'hardware e il programmatore o tra software a basso e quello ad alto livello semplificando così il lavoro di programmazione.
5, 15

Bibliography

Article and papers

- [1] Lorenzo Perinello. Alberto Morini Claudio Enrico Palazzi. *GoodIT 2025 - International Conference on Information Technology for Social Good*. 2025. URL: <https://goodit2025.org> (cit. on p. ii).
- [3] Leon Ciechanowski Amin Mahmoudi Dariusz Jemielniak. *Echo Chambers in Online Social Networks: A Systematic Literature Review*. 2024. URL: <https://ieeexplore.ieee.org/abstract/document/10388309> (cit. on p. 2).
- [6] Aoxia Chen and Pankaj K. Sen. *Advancement in battery technology: A state-of-the-art review*. 2016. URL: <https://ieeexplore.ieee.org/document/7731812> (cit. on p. 1).
- [8] Ophir Frieder. Eugene Yang David D. Lewis. *Social Network Sites: Definition, History, and Scholarship Open Access*. 2021. URL: <https://arxiv.org/abs/2108.12752> (cit. on p. 2).
- [9] Giulia Fioravanti et al. *Fear of missing out and social networking sites use and abuse A meta-analysis*. 2021. URL: <https://www.sciencedirect.com/science/article/abs/pii/S074756322100162X> (cit. on p. 4).
- [26] Yuhao Zhu Matthew Halpern and Vijay Janapa Reddi. *Mobile CPU's rise to power: Quantifying the impact of generational mobile CPU design trends on performance, energy, and user satisfaction*. 2016. URL: <https://ieeexplore.ieee.org/document/7446054> (cit. on p. 1).
- [27] Hany Farid. Matyáš Boháček. *Protecting President Zelenskyy against Deep Fakes*. 2022. URL: <https://arxiv.org/abs/2206.12043> (cit. on p. 2).
- [28] Zahra Khanbabaloo. Mohammad Hajarian. *Toward Stopping Incel Rebellion: Detecting Incels in Social Media Using Sentiment Analysis*. 2021. URL: https://www.researchgate.net/profile/Mohammad-Hajarian/publication/352100924_Toward_Stopping_Incel_Rebellion_Detecting_Incels_in_Social_Media_Using_Sentiment_Analysis/links/60b8e34992851cb13d73e858/Toward-Stopping-Incel-Rebellion-Detecting-Incels-in-Social-Media-Using-Sentiment-Analysis.pdf (cit. on p. 2).
- [34] Andrew Tomk. Ravi Kum Jasmine Novak. *Structure and Evolution of Online Social Networks*. 2006. URL: <https://dl.acm.org/doi/abs/10.1145/1150402.1150476> (cit. on p. 2).
- [36] Dmitri Rozgonjuk et al. *Fear of Missing Out (FoMO) and social media's impact on daily life and productivity at work: Do WhatsApp, Facebook, Instagram, and Snapchat use disorders mediate that association?* 2020. URL: <https://www.sciencedirect.com/science/article/abs/pii/S0306460320306171> (cit. on p. 4).

Web sites

- [2] Anitha Edison Amala Mary. *Deep fake Detection using deep learning techniques A Literature Review*. 2023. URL: <https://ieeexplore.ieee.org/abstract/document/10164881/authors#authors> (cit. on p. 2).
- [4] Shariq Aziz Butt et al. *Remote mobile health monitoring frameworks and mobile applications: Taxonomy, open challenges, motivation, and recommendations*. 2024. URL: <https://www.sciencedirect.com/science/article/abs/pii/S0952197624003919> (cit. on p. 1).
- [5] Windows Central. *Windows Phone isn't dead, Part VI: App Gap? Microsoft has a platform for that*. 2016. URL: <https://www.windowscentral.com/windows-phone-isnt-dead-part-vi-app-gap> (cit. on p. 1).
- [7] Tech Crunch. *Instagram rolls out a new searchable map to make it easier to discover popular locations*. 2022. URL: <https://techcrunch.com/2022/07/19/instagram-new-searchable-map-experience/> (cit. on p. 5).
- [10] Waze Mobile Ltd. Google. *Waze official page*. 2006. URL: <https://www.waze.com/apps> (cit. on p. 1).
- [11] Jeffrey Gottfried. *Americans' Social Media Use: YouTube and Facebook are by far the most used online platforms among U.S. adults; TikTok's user base has grown since 2021*. 2024. URL: <https://www.pewresearch.org/internet/2024/01/31/americans-social-media-use/> (cit. on p. 2).
- [12] Apple Inc. *iOS official page*. 2007. URL: <https://www.apple.com/os/ios/> (cit. on p. 1).
- [13] Google Inc. *Android official page*. 2008. URL: <https://www.android.com/> (cit. on p. 1).
- [14] Google Inc. *Google Lens official page*. 2017. URL: <https://lens.google/> (cit. on p. 1).
- [15] Google Inc. *Google Maps official page*. 2005. URL: <https://www.google.com/maps/about/> (cit. on p. 1).
- [16] Google Inc. *Google shuts failed social network Google*. 2019. URL: <https://www.bbc.com/news/technology-47771927> (cit. on p. 3).
- [17] Google Inc. *See where your friends are with Google Latitude*. 2009. URL: <https://googleblog.blogspot.com/2009/02/see-where-your-friends-are-with-google.html> (cit. on p. 3).
- [18] Meta Inc. *Meta Reports Fourth Quarter and Full Year 2023 Results; Initiates Quarterly Dividend*. 2024. URL: <https://investor.atmeta.com/investor-news/press-release-details/2024/Meta-Reports-Fourth-Quarter-and-Full-Year-2023-Results-Initiates-Quarterly-Dividend/default.aspx> (cit. on p. 2).
- [19] Meta Inc. *Scaling the Instagram Explore recommendations system*. 2023. URL: <https://engineering.fb.com/2023/08/09/ml-applications/scaling-instagram-explore-recommendations-system/> (cit. on p. 5).
- [20] Snap Inc. *Snap Inc. Announces Third Quarter 2025 Financial Results*. 2025. URL: <https://investor.snap.com/news/news-details/2025/Snap-Inc--Announces-Third-Quarter-2025-Financial-Results/default.aspx> (cit. on p. 4).
- [21] Snap Inc. *Snapchat official page*. 2011. URL: <https://www.snapchat.com/> (cit. on p. 4).

- [22] Marina Niessner, J. Anthony Cookson William Mullins. *Social Media and Finance*. 2024. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4806692 (cit. on p. 2).
- [23] Steven Furnell, Karl van der Schyff Stephen Flowerday. *Duplicitous social media and data surveillance An evaluation of privacy risk*. 2020. URL: <https://www.sciencedirect.com/science/article/abs/pii/S0167404820300961> (cit. on p. 2).
- [24] Andrei O. J. Kwok and Sharon G. M. Koh. *Deepfake: a social construction of technology perspective*. 2020. URL: <https://www.tandfonline.com/doi/abs/10.1080/13683500.2020.1738357> (cit. on p. 2).
- [25] Peter James Marcin Straczekiewicz and Jukka-Pekka Onnela. *A systematic review of smartphone-based human activity recognition methods for health research*. 2021. URL: <https://www.nature.com/articles/s41746-021-00514-4> (cit. on p. 1).
- [29] ABC News. *Snapchat's new Snap Map feature raises privacy concerns*. 2017. URL: <https://abcnews.com/Lifestyle/snapchats-snap-map-feature-raises-privacy-concerns/story?id=48271889> (cit. on p. 4).
- [30] Neliia Blynova Oksana Kyrylova and Viktoriia Pavlenko. *Mobile internet usage world-wide - statistics and facts*. 2023. URL: <https://www.statista.com/topics/779/mobile-internet/>.
- [31] Neliia Blynova Oksana Kyrylova and Viktoriia Pavlenko. *The perspectives for mobile application use in media education*. 2023. URL: <https://www.tandfonline.com/doi/full/10.1080/10494820.2023.2186897> (cit. on p. 1).
- [32] A. Perrin. *Social Media Usage: 2005–2015 – 65% of adults now use social networking sites, a nearly tenfold jump in the past decade*. 2015. URL: https://www.secretintelligenceservice.org/wp-content/uploads/2016/02/PI_2015-10-08_Social-Networking-Usage-2005-2015_FINAL.pdf (cit. on p. 2).
- [33] Amanda Raffoul et al. *Social media platforms generate billions of dollars in revenue from U.S. youth Findings from a simulated revenue model*. 2023. URL: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0295337> (cit. on p. 2).
- [35] Reuters. *Meta CEO Zuckerberg says Instagram has grown to 3 billion monthly active users*. 2025. URL: <https://www.reuters.com/business/meta-ceo-zuckerberg-says-instagram-has-grown-3-billion-monthly-active-users-2025-09-24/> (cit. on p. 5).
- [37] Apple Inc. Shazam Entertainment Limited. *Shazam official page*. 2002. URL: <https://www.shazam.com/> (cit. on p. 1).
- [38] Statista. *Device usage of Facebook users worldwide as of January 2022*. 2022. URL: <https://www.statista.com/statistics/377808/distribution-of-facebook-users-by-device/> (cit. on p. 1).
- [39] Statista. *Global smartphone sales to end users from 2007 to 2023*. 2023. URL: <https://www.statista.com/statistics/263437/global-smartphone-sales-to-end-users-since-2007/> (cit. on p. 1).
- [40] Statista. *M Number of monthly active Instagram users from January 2013 to December 2021*. 2021. URL: <https://www.statista.com/statistics/253577/number-of-monthly-active-instagram-users/> (cit. on p. 5).

- [41] Statista. *Most popular global mobile messenger apps as of October 2025, based on number of monthly active users*. 2025. URL: <https://www.statista.com/statistics/258749/most-popular-global-mobile-messenger-apps/> (cit. on p. 1).
- [42] Social Media Today. *Instagram Stories is Now Being Used by 500 Million People Daily*. 2019. URL: <https://www.socialmediatoday.com/news/instagram-stories-is-now-being-used-by-500-million-people-daily/547270/> (cit. on p. 5).
- [43] Eastern District of New York. U.S. Attorney’s Office. *Two Individuals Arrested for Publishing AI Deepfake Pornography In Violation Of TAKE IT DOWN Act*. 2026. URL: <https://www.justice.gov/usao-edny/pr/two-individuals-arrested-publishing-ai-deepfake-pornography-violation-take-it-down-act> (cit. on p. 2).
- [44] International Telecommunication Union. *International Telecommunication Union. 2023. Facts and Figures 2023 – Mobile Phone Ownership*. 2023. URL: <https://www.itu.int/itu-d/reports/statistics/2023/10/10/ff23-mobile-phone-ownership/> (cit. on p. 1).
- [45] Computer World. *Windows Phone still has serious app gap despite Microsoft’s \$7.2 billion Nokia buyout*. 2013. URL: <https://www.computerworld.com/article/1500308/windows-phone-still-has-serious-app-gap-despite-microsoft-s-7-2-billion-nokia-buyo.html> (cit. on p. 1).